ZCAS UNIVERSITY

BScCS41 - BACHELOR OF SCIENCE IN CYBER SECURITY/BScNE41- NETWORK ENGINEERING

CIT4351 CCNA CYBERSECURITY OPERATIONS

FINAL EXAMINATIONS

TUESDAY, 26th MAY 2026

TIME: 14:00 – 17:00 HOURS

TIME ALLOWED: THREE (03) HOURS

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is **compulsory** with **ONE (01)** Question and carries **40 marks.**
2. **SECTION B:** Attempt **any THREE (03)** questions from this section. Each question carries **20 marks.**
3. This examination paper carries a total of **100 marks.**
4. Time allowed: **THREE (03) Hours plus FIVE (05) minutes to read through the paper.**
5. **DO NOT** turn this page **until** you are told to do so.

SECTION A: Question ONE is COMPULSORY

QUESTION ONE

SCENARIO

LusakaPay Technologies is a fast-growing mobile money and fintech company operating across Zambia. The company recently experienced a coordinated cyberattack. Employees were targeted with spear-phishing emails impersonating the Ministry of Finance. Shortly after, attackers deployed ransomware that encrypted internal accounting systems while quietly extracting customer KYC data to an external server. Security monitoring was ineffective because logs were not centralized, and SIEM alerts were not actively investigated. The breach was only discovered after customers reported failed transactions and account discrepancies.

The management now intends to implement a Security Operations Center (SOC) with strong Threat Intelligence capabilities.

Assume you are the SOC Lead Analyst at LusakaPay Technologies.

a) Explain the importance of cybersecurity policies in LusakaPay Technologies and how they support proactive defense mechanisms. (08 marks)

b) Identify and describe four (04) possible attack vectors used in this breach. (08 marks)

c) Describe four (04) core SOC security tools that would improve detection and response at LusakaPay Technologies. (08 marks)

d) Differentiate between IDS and IPS in terms of detection method and response capability. (08 marks)

e) Explain the CIA Triad and its relevance in protecting financial systems like LusakaPay Technologies. (04 marks)

f) Explain how SIEM systems and log correlation could have helped detect the attack earlier. (04 marks)

(Total: 40 marks)

SECTION B - CHOOSE ANY THREE (03) QUESTIONS

QUESTION TWO

SCENARIO

ZAMTEL Data Services detected unusual spikes in DNS requests from a corporate client. Further investigation revealed DNS tunneling used to exfiltrate sensitive billing data in encoded form. The attack went unnoticed for weeks due to lack of proper DNS monitoring and anomaly detection systems.

a) Define DNS tunneling and explain why attackers use it. (04 marks)

b) Identify four (04) indicators of DNS tunneling in network traffic logs. (04 marks)

c) Recommend two (02) tools used for detecting DNS anomalies. (04 marks)

d) Suggest two (02) preventive network controls against DNS-based attacks. (04 marks)

e) Identify TWO (02) SOC roles responsible for handling DNS-based threats. (04 marks)

(Total: 20 Marks)

QUESTION THREE

As a SOC analyst, you receive the following simplified log entries:

02:14:33 VPN Login Success
User: j.mwale
Source IP: 185.203.116.45
Location: Unknown

02:15:10 Database Access
User: j.mwale
File Accessed: energy_clients_backup.sql

02:18:27 Firewall Alert
Port Scan Detected
Source: 185.203.116.45
Destination: Internal Network

02:22:05 File Execution Alert
Host: ENG-LAPTOP-07
File: financial_report_2026.pdf.exe

Required:

a) Identify **four (04) suspicious indicators.** (04 marks)

b) Explain what each log entry suggests. (04 marks)

c) Determine whether the incident indicates **account compromise or insider threat.** (04 marks)

d) Identify FOUR Linux commands the analyst should use to investigate this activity (e.g., listing processes, checking open network connections, reviewing logs), and explain what information each command provides. (04 marks)

(Total: 20 Marks)

QUESTION FOUR

SCENARIO

AgroTrade Zambia uses cloud services such as Microsoft 365 and Azure for agricultural supply chain management. A misconfigured OneDrive sharing policy exposed sensitive export contracts and employee payroll data to the public internet. The breach was discovered when a third party indexed the files on a search engine. No audit logging or Data Loss Prevention (DLP) policies were enabled.

a) Recommend FOUR (04) user awareness training strategies to prevent such breaches. (04 marks)

b) Explain the Zero Trust security model and its application in cloud environments like AgroTrade Zambia. (04 marks)

c) Identify TWO (02) Microsoft security tools you can use for cloud threat detection. (04 marks)

d) Differentiate between full packet capture and metadata analysis. (04 marks)

e) Explain the role of NetFlow in monitoring network traffic. (04 marks)

(Total: 20 Marks)

QUESTION FIVE

SCENARIO

Copperbelt Energy Systems experienced a ransomware attack targeting both its billing infrastructure and SCADA industrial control systems. Attackers used spear-phishing emails disguised as regulatory compliance notices from ZESCO. Systems were encrypted and a ransom demanded in cryptocurrency.

The organization lacks a structured Cyber Threat Intelligence (CTI) program and has no formal threat-sharing partnerships.

a) Define Cyber Threat Intelligence (CTI) and explain its importance in protecting critical infrastructure. (04 marks)

b) Describe the FOUR (04) types of threat intelligence: strategic, tactical, operational, and technical. (04 marks)

c) Explain how MITRE ATT&CK or the Cyber Kill Chain can help in detecting and mitigating attacks. (04 marks)

d) Identify FOUR (04) types of cyber threat actors. (04 marks)

e) Distinguish between cybersecurity and cybersecurity operations. (04 marks)

(Total: 20 marks)

END OF EXAMINATIONS